Custom Parametric Shoe Trees

By Alex, James and Naijun

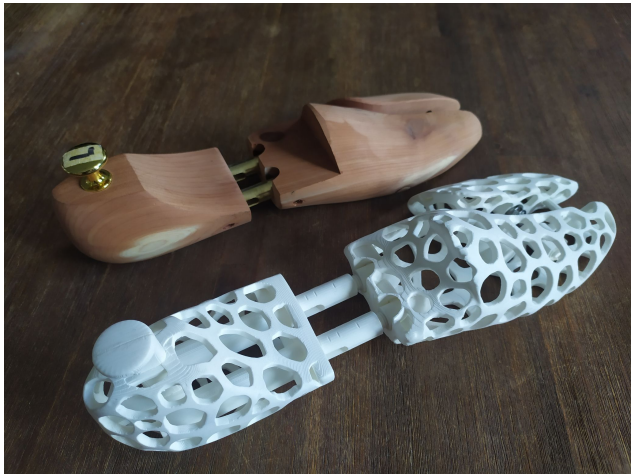
Problem Statement

Traditional shoe trees are designed with "universal" sizes that only expand at fixed points like the toe box or length.

This mismatch causes shoes to deform in ways that don't align with the user’s actual foot, leading to permanent structural damage and unnecessary wear.

Furthermore, commercial versions are significantly overpriced at $20–$30, whereas a custom, breathable 3D-printed alternative can be produced for just a few dollars.

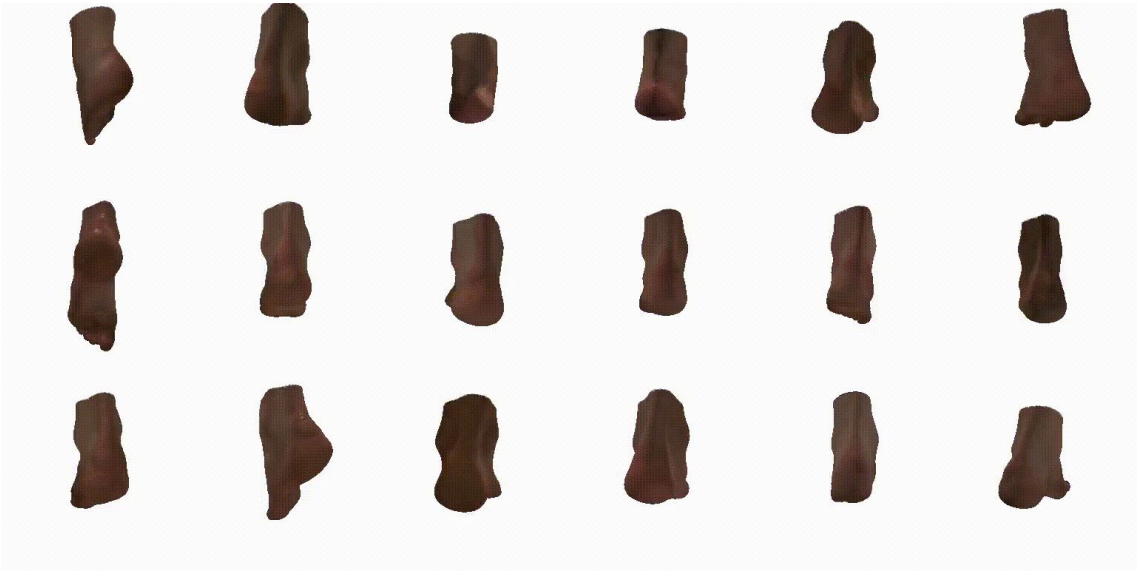
Athletic shoes are especially succeptibule to deformation after use, customization of shoetrees targets this issue and helps speed up the break in process



Related Work

A high-quality collection of textured 3D foot scans

Provides a rigorous testing data for alignment and measurement algorithm

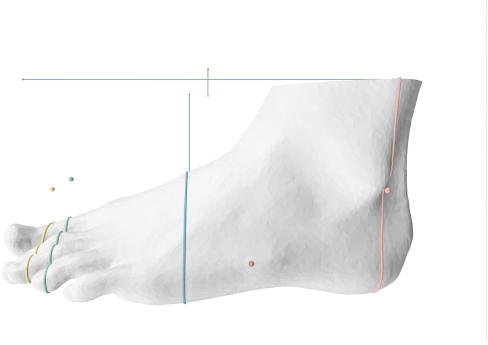


https://www.ollieboyne.com/FIND

Overall Pipeline

Language: Python

- **Phase 1: Input Normalization**
  - Coordinate alignment, unit scaling, and feature extraction.
- **Phase 2: Template Registration**
  - FPFH + ICP matching to a canonical shoe last model.
- **Phase 3: Parametric Warping**
  - RBF-based deformation driven by measured foot dimensions.
- **Phase 4: Geometry Refinement**
  - Toe-aware "sock-wrap" cleanup and mesh smoothing.
- **Phase 5: Final Export**
  - Watertight STL generation and print-volume analysis

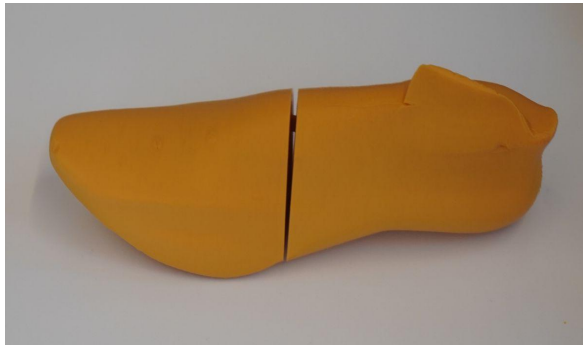


Phase 1

Phase 2

Phase 3

Phase 4



Phase 5

Alignment

The Challenge: Inconsistent Inputs

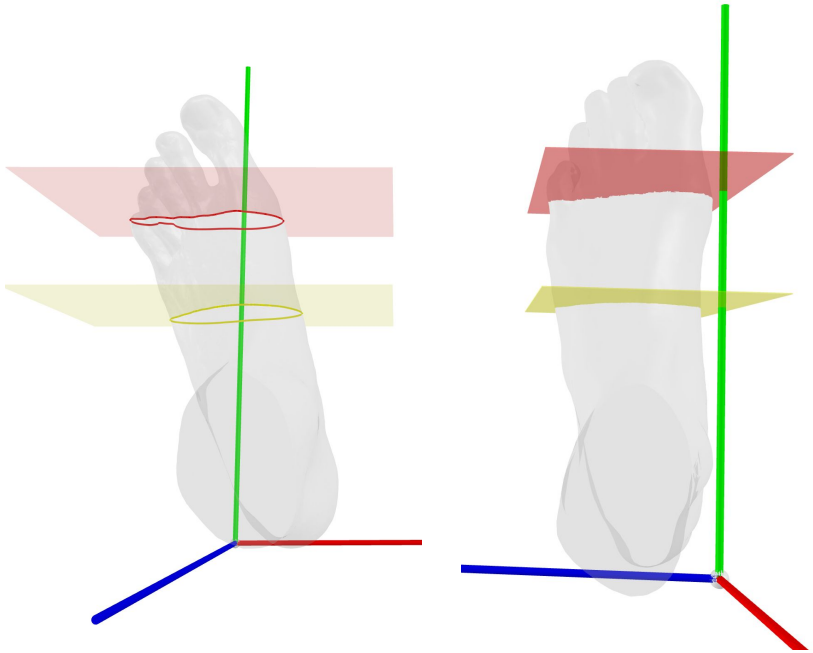
- **Stochastic Data:** Raw scans lack a unified coordinate system.
- **Risk:** Varied orientations and scales lead to **inaccuracy** in extracting measurements and warping.

Stage 1: Coarse Alignment

- **Normalization:** Auto-scales units to mm.
- **Semantic Correction:** Uses **Sectional Area** and **Thickness Gradients** to identify sole/toe.
- **Role:** Corrects "upside down" or "backward" poses.

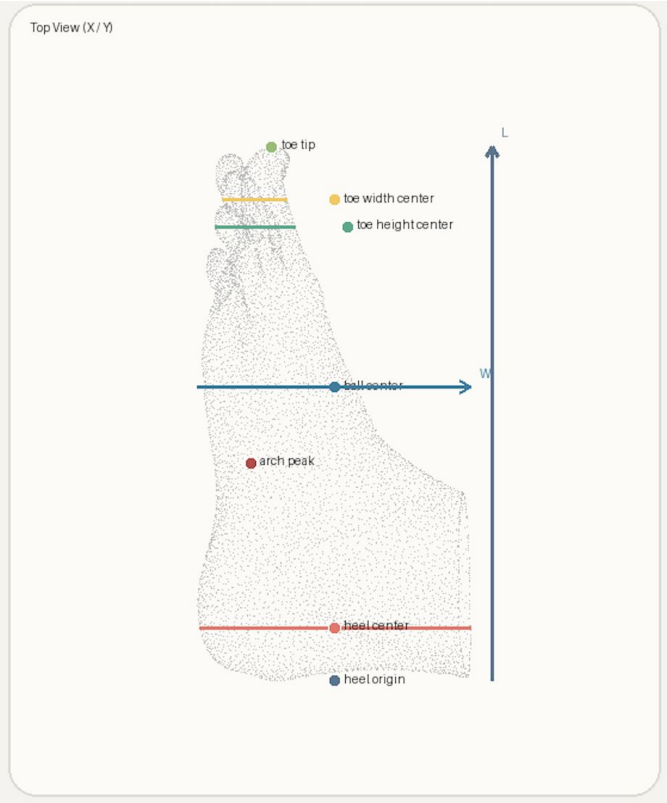
Stage 2: Fine Alignment

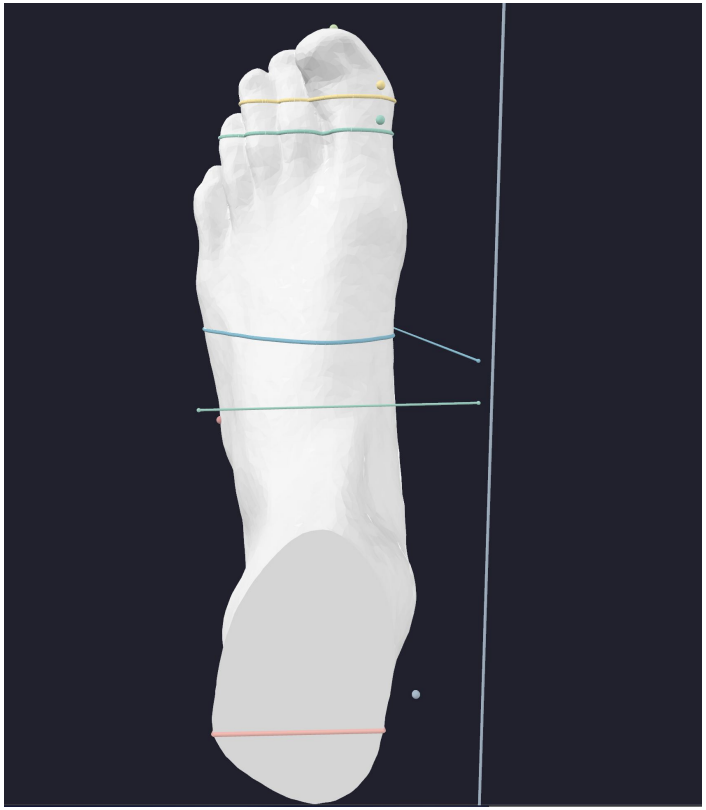
- **SVD Plane Fitting:** Fits a precision plane to the lowest 8% of vertices.
- **Canonical Snapping:** Locks heel to Y=0 and sole to Z=0.
- **Role:** Finalizes coordinates for sub-millimeter accuracy.



Before alignment After alignment

Measurement





Warping & Modeling

Dynamic Profile Sampling:

- Slices the template into **60 segments** to build a high-resolution geometric profile.

Adaptive Control Points:

- Instead of global scaling, it uses **7 width-control points** and **8 height-control points** (heel, ball, arch, toe).

Mathematical Deformation (Gaussian Smoothing):

- Uses **1D Gaussian filters** to interpolate between control points, ensuring a smooth, organic transition across the shoe last.

Geometry Safety (Collar Fin Trimming):

- Detects and removes degenerate faces (fins) caused by extreme scaling to ensure a **manifold and printable mesh.**

Trimming and Splitting

Mesh Cleaning:

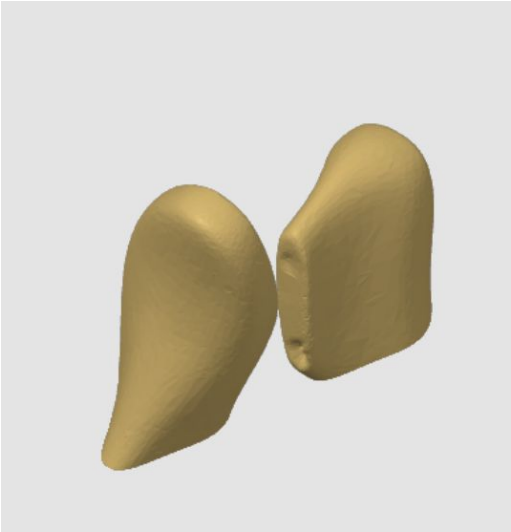
- Repairs holes and non-manifold edges in the input STL so boolean operations can run without error (pymeshfix)

Model Split:

- Cuts the cleaned mesh into two halves along the Y axis at the midpoint, so each half fits within the printer bed limits (trimesh)

Create Clips for strong connection:

- Generates parametric snap-fit tabs on one half and matching sockets on the other, so the two halves lock together after printing. (shapely)

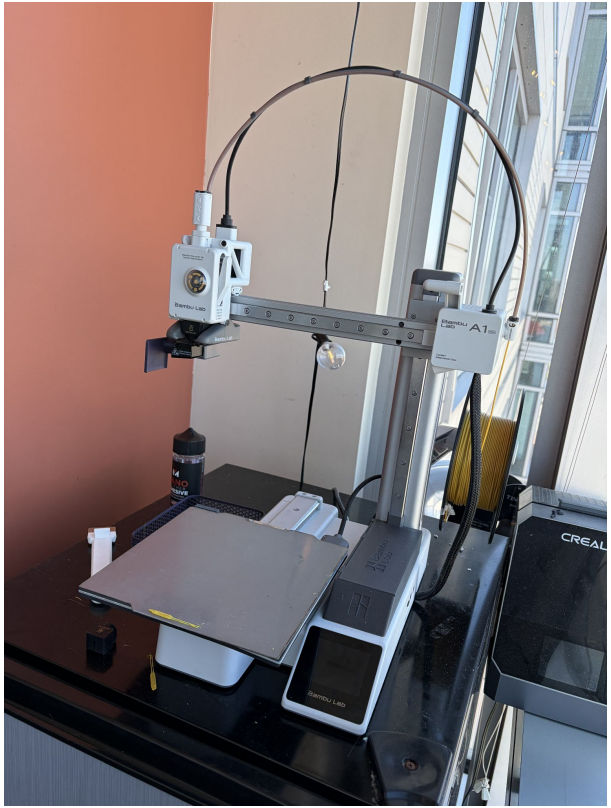


Fabrication

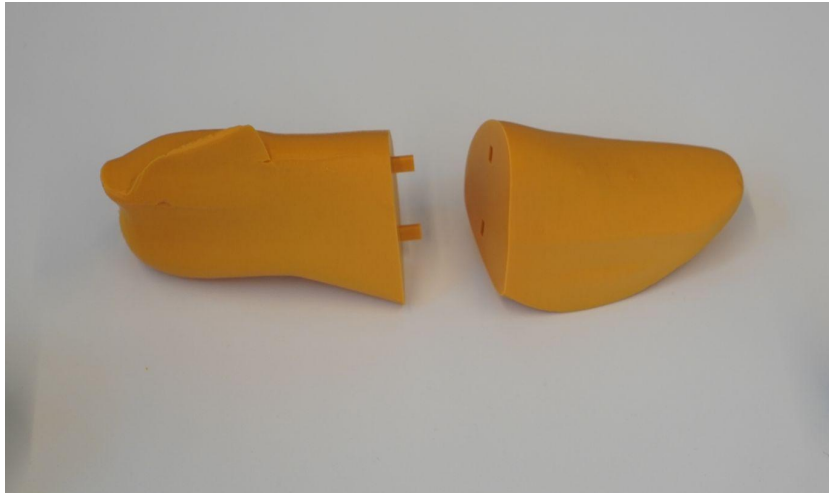
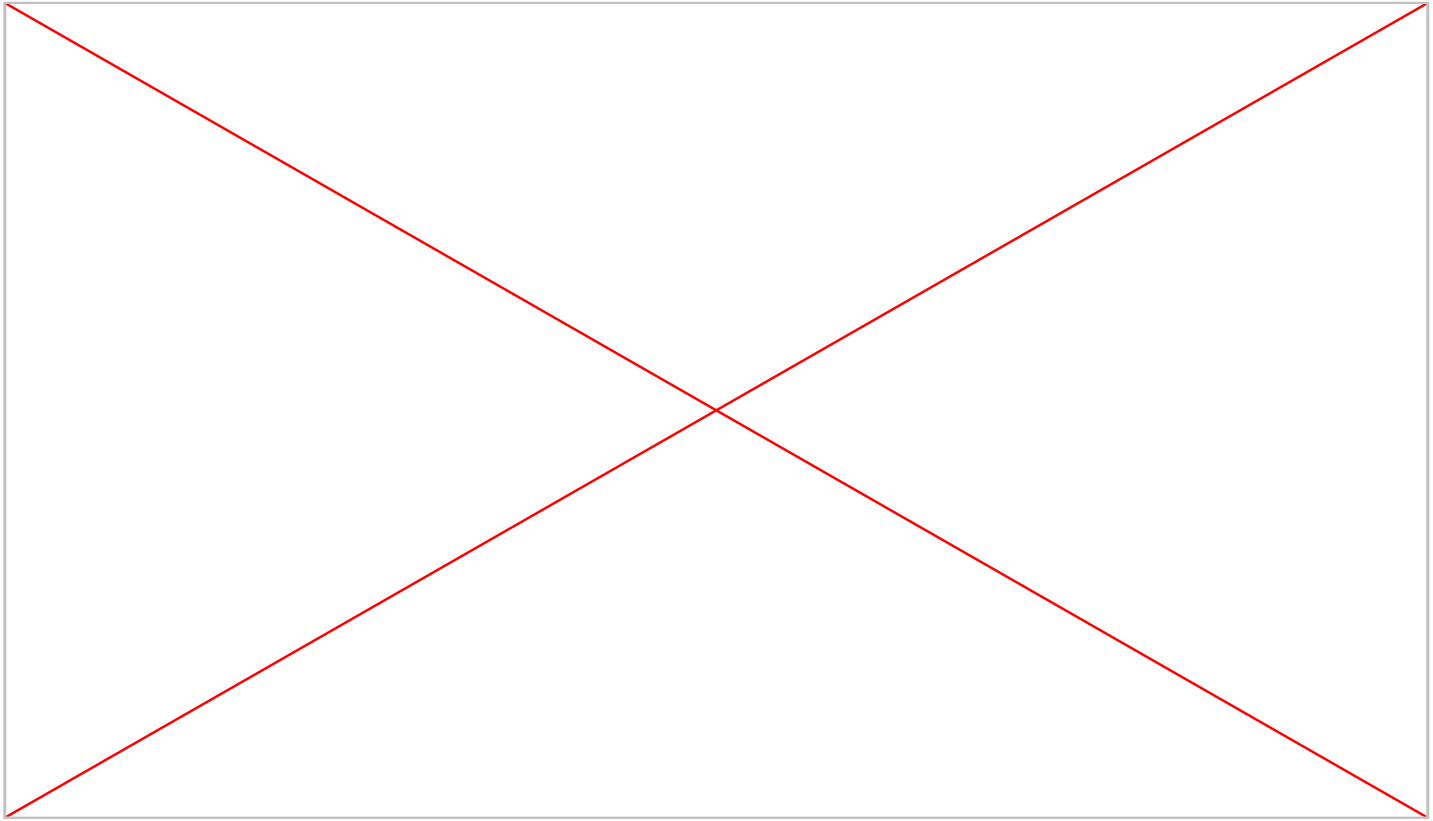
Print Overall

- The largest size 13 shoe tree to 6.5 hours to print @ 5% infill
- Minimal challenges since the shape is simple aside from printer calibration issues that had to be handled

Smaller prints were done split side down for the best quality and prints took around 3 hours



Result



Result